Title : Project Registration & Progress Review **FF No. 180**

Department: E & Tc	**Academic Year: 2026-27**
Semester: I	**Group No.: 22**
Project Title: Fruit shelf life estimation for Climacteric fruits	
Project Area: Embedded , ML And Computer vision	

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Project approved / Not approved

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Project Synopsis :-

a. Introduction :

We all have experience of Indian fruit market where we brought after guesswork self sorting we don't get the proper quality of fruits. Also farmers and big market reservoirs also don't have techniques to sort out fruits based on their remaining shelf life. If they sort out fruits based on theis shelf by using some scientific techniques it will be beneficial for both farmers and market leaders also to manage the fruit marketing. eliminating guesswork and reducing post-harvest losses that cost Indian farmers an estimated Rs 90,000 crore annually.

Key Features :- NIR Sensor data model, Spectral analysis

Aim : To provide low cost solution to estimate fruit shelf life typically for climacteric fruits

b. Problem Statement :

Traditional fruit freshness assessment is often subjective, destructive, and time-consuming. This project aims to develop a non-destructive and cost-effective system using NIR sensing and camera-based spectral analysis to detect fruit ripeness, assess quality, and predict its remaining shelf life using machine learning.

c. Objectives :

1) Analyze the first individual fruits NIR data
2) Perform Spectral analysis using normal cameras
3) Determine the shelf life of fruit using both factors

d. Literature Review :

Near infrared spectroscopy (700-2500nm wavelength range) is the industry standard for non-destructive fruit quality assessment. Key chemical bonds in fruit absorb NIR light at specific wavelengths:

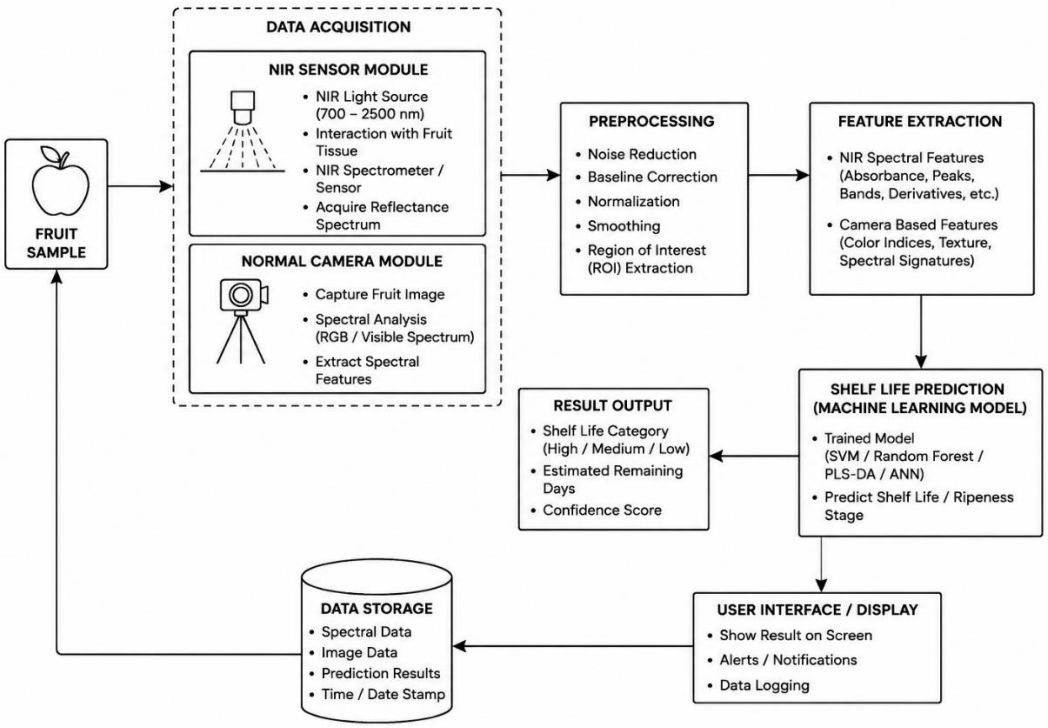
- Sugar (Brix content): Glucose and fructose absorb at 1450nm and 2100nm
- Water content: Strong absorption at 970nm and 1450nm
- Starch to sugar conversion: Shift in absorbance at 850-1000nm as ripening progresses
- Lycopene (tomato ripeness): Absorbance shift at 475nm (visible) and 700nm

Recent research in agricultural engineering focuses on non-destructive techniques for fruit freshness, ripeness, and shelf-life prediction. Mahata et al. (2023) used VOC sensors with machine learning to estimate fruit freshness and storage duration, while Zhao et al. (2024) combined artificial olfactory sensing with deep learning for automated ripeness assessment.

Near-Infrared (NIR) spectroscopy has also emerged as an effective method for evaluating internal fruit quality without damaging the sample. Wang et al. (2025) highlighted Vis/NIR spectroscopy for quality assessment, while Amodio et al. (2025) demonstrated portable NIR spectroscopy with machine learning for accurate ripening-stage classification.

For external quality assessment, RGB sensors and computer vision provide cost-effective solutions. Pawar et al. (2022) used an RGB color sensor with logistic regression for ripeness classification, while Noutfia et al. (2024) demonstrated camera-based monitoring of external quality. Deep-learning approaches such as ResNet-50 (Ban et al., 2023) and YOLOv4 (Cheng et al., 2022) further improved maturity detection and grading accuracy. Overall, combining NIR sensing, RGB imaging, and machine learning provides a promising framework for non-destructive fruit shelf-life prediction.

e. System Architecture :-



Hardware Components :-

Sensor	Wavelength	Interface	Use Case
TEMT6000	Visible 570 nm peak	Analog	Light levels only, not NIR
BPW34 photodiode + NIR LED	550–1100 nm	Analog	DIY single-band reflectance
AS7265x (Adafruit)	410–940 nm, 18 channels	I2C	Full spectral profile
Raspberry Pi NoIR + 850 nm filter	750–1000 nm image	CSI	Spatial ripeness mapping
MLX90614 IR temp	5.5–14 µm (thermal)	I2C	Surface temperature proxy for ripeness

f. References

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2) Ban, Z., Fang, C., Liu, L., Wu, Z., Chen, C., & Zhu, Y. (2023). Detection of Fundamental Quality Traits of Winter Jujube Based on Computer Vision and Deep Learning. Agronomy, 13, 2095.
3) Cheng, Y.-H., Tseng, C.-Y., Nguyen, D.-M., & Lin, Y.-D. (2022). YOLOv4-Driven Appearance Grading Filing Mechanism: Toward a High-Accuracy Tomato Grading Model through a Deep-Learning Framework. Mathematics, 10, 3398.
4) Mahata, B., Acharyya, S., Giri, S., Mahata, T., Banerji, P., & Guha, P. K. (2023). Fruit Freshness Monitoring Employing Chemiresistive Volatile Organic Compound Sensor and Machine Learning. ACS Applied Nano Materials, 6, 22829–22836.
5) Noutfia, Y., Ropelewska, E., Jóźwiak, Z., & Rutkowski, K. (2024). Non-Destructive Monitoring of External Quality of Date Palm Fruit (Phoenix dactylifera L.) During Frozen Storage Using Digital Camera and Flatbed Scanner. Sensors, 24, 7560.
6) Pawar, S., Meshram, J., Mondhe, A., Nachan, A., Nilangekar, T., & Patil, A. (2022). Ripeness Detector for Vegetables and Fruits. 2022 2nd International Conference on Intelligent Technologies (CONIT), Hubli, India, pp. 1-5.
7) Wang, X., et al. (2025). Nondestructive Quality Detection of Characteristic Fruits Based on Vis/NIR Spectroscopy: Principles, Systems, and Applications. Agriculture, 15(20), 2167.
8) Zhao, M., You, Z., Chen, H., Wang, X., Ying, Y., & Wang, Y. (2024). Integrated Fruit Ripeness Assessment System Based on an Artificial Olfactory Sensor and Deep Learning. Foods, 13, 793.

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Group No.	22		
Activity	Review Schedule	Progress Review Report submitted	Signature of Guide
Review 1	Mid Sem. Semester	Yes / No	
Review 2	End of Semester	Yes / No	

Format of Progress Review Report:

Review No.: 1 **Group No.: 22** **Date: 23/09/26**

Progress Review Report

Title: Fruit shelf life estimation for Climacteric fruits

Literature Review

1. Lauretti et al. (2025), IEEE Sensors Journal

Positions itself against hyperspectral cameras (need controlled lighting), low-cost multispectral sensors with hand-crafted fruit-specific indices (not generalizable), and RGB classifiers (lower accuracy).

- ***Components:*** **Roithner SMB1N-BB450 broadband LED (340–1050nm); two Hamamatsu sensors (C12880MA + C14384MA) giving 52 channels; custom electronics (5V, 235mW idle/740mW active); 3D-printed dark enclosure; <€600 total.**
- ***Methodology:*** **Moving-average filter → normalize $R_n=(R-D)/(W-D)$ → ReliefF feature selection → linear SVM, 10-fold CV. Tested on 450 tomato scans, 6 ripeness stages.**
- ***Result:*** **93.72% (52-channel) vs. 88.52% (18-channel) vs. 86.92% (RGB).**

2. Zbotic.in

- ***Components:*** **850/940nm NIR LED, BPW34/SFH213 photodiode, MCP6001 transimpedance amp, Arduino Uno/Nano, dark chamber; optional AS7265x (18-channel) for richer data.**
- ***Methodology:*** **Dark reading (LED off) + illuminated reading (LED on) → relative reflectance % vs. white reference → threshold into unripe/nearly ripe/ripe/overripe. Thresholds must be recalibrated per fruit variety.**
- ***Claimed accuracy:*** **80–90% (DIY) vs. 95–98% (commercial spectrophotometer).**

3. RipeTrack (Waseem, Sharma, Hefeeda), IEEE TMC

Contrasts with ethylene sensors, benchtop NIR spectrometers, and sub-THz sensing — all needing specialized hardware.

- ***Approach:*** **Reconstructs full 400–1000nm hyperspectral-like signatures from ordinary smartphone RGB+NIR images via a vision-transformer reconstruction model, then classifies ripeness/remaining lifetime with a spatio-spectral classifier.**
- ***Validation:*** **7 fruits, 1,913 hyperspectral images, 1,144 ethylene readings (ground truth), 3,865 phone image pairs, tested in 5 real grocery stores.**
- ***Result:*** **≥96% ripeness accuracy, ≥93% remaining-lifetime accuracy on unmodified**

phones under diverse lighting.

4. Adafruit AS7343 14-Channel Light/Color Sensor Breakout

- **AS7343 sensor IC: 25 photodiodes covering 11 overlapping visible bands + 1 NIR + 1 clear + 1 flicker channel (14 channels total; 18 raw readings across 3 SMUX cycles), routed through a 16-bit 6-channel ADC**
- **Onboard white LED (driven directly by the AS7343's built-in LED driver) for active illumination of a target sample**
- **STEMMA QT / Qwiic I2C connectors (fixed address 0x39), onboard 3–5V→3V regulator, INT and GPIO pins**
- **Wavelength channels: 405, 425, 450, 475, 515, 550, 555, 600, 640, 690, 745, 855 nm, plus clear and flicker-detection channels**

Components

1. **Adafruit AS7343 14-Channel Light / Color Sensor Breakout**
2. **Arduino UNO**
3. **3D-printed dark enclosure that contacts the fruit to block ambient light**

Methodology:

1. **Establish that ripening-related spectral changes in bananas are detectable directly in the 405–855 nm range using the AS7343's 14 built-in channels — no reconstruction step needed, since the sensor natively captures 12 spectral bands + clear/flicker channels rather than just RGB, unlike a phone camera.**
2. **Since the AS7343 already outputs discrete spectral channels (not a 3-band RGB image), skip the vision-transformer upscaling RipeTrack needs — read all 18 raw channel values per sample directly via readAllChannels() over I2C, with the onboard white LED as the illumination source and gain/ATIME/ASTEP fixed for consistency across readings.**
3. **Define ground-truth ripeness labels by day rather than by ethylene emission curve (no gas sensor in this setup): Day 1–2 = Medium, Day 3–5 = Ripe, Day 6–7 = Overripe, following the same conceptual unripe→ripe→expired progression RipeTrack uses.**
4. **Feed the 12-channel spectral readings per sample directly into a classifier (planned: SVM or similar, echoing Lauretti et al.'s pipeline) to predict ripeness stage, rather than RipeTrack's spatio-spectral attention network — a much lighter model suited to 12 features instead of 68 reconstructed bands.**
5. **Validated (so far) on 5 bananas tracked across 7 days, 2 readings per sample per day (70 rows total) — a synthetic dataset generated from your original Day 1 sensor readings using a per-channel decay model, rather than real daily AS7343 scans or hyperspectral ground truth.**
6. **Next steps to match RipeTrack's validation rigor: replace the synthetic decay model with real AS7343 readings taken daily on the actual 5 bananas, add a white/dark reference calibration per the AS7343 setup discussed earlier, and eventually test classification accuracy against the true ripeness stage rather than an assumed day-based label.**

Signature of Guide:

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Review No.: 2	**Group No.:**	**Date:**
Progress Review Report		
Signature of Guide:		